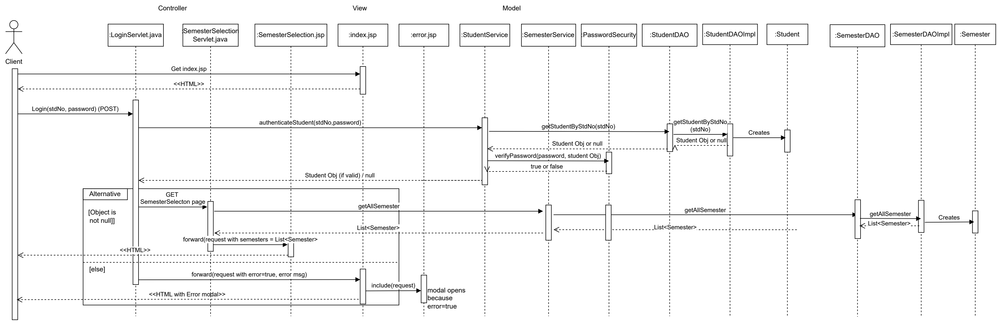
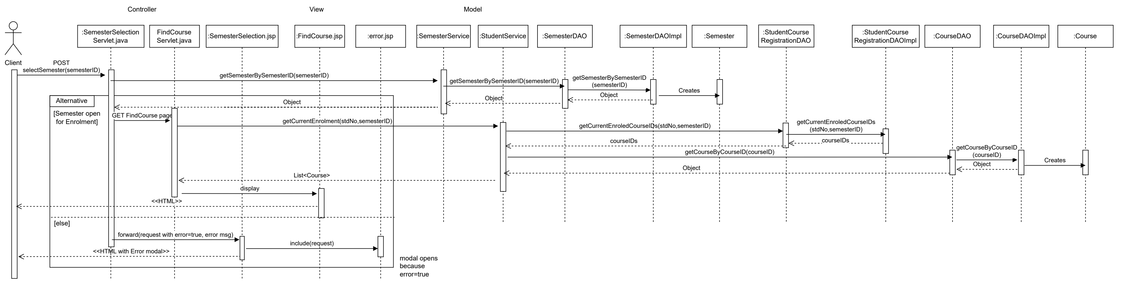
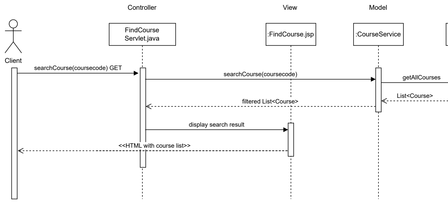
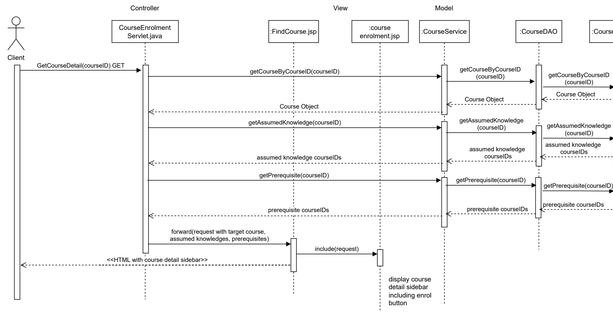
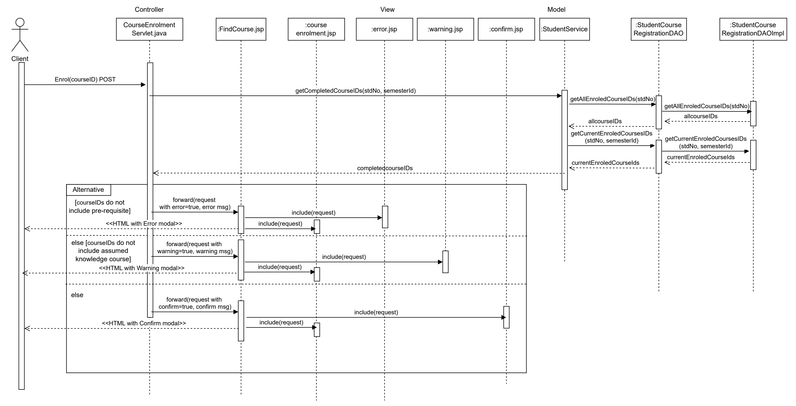
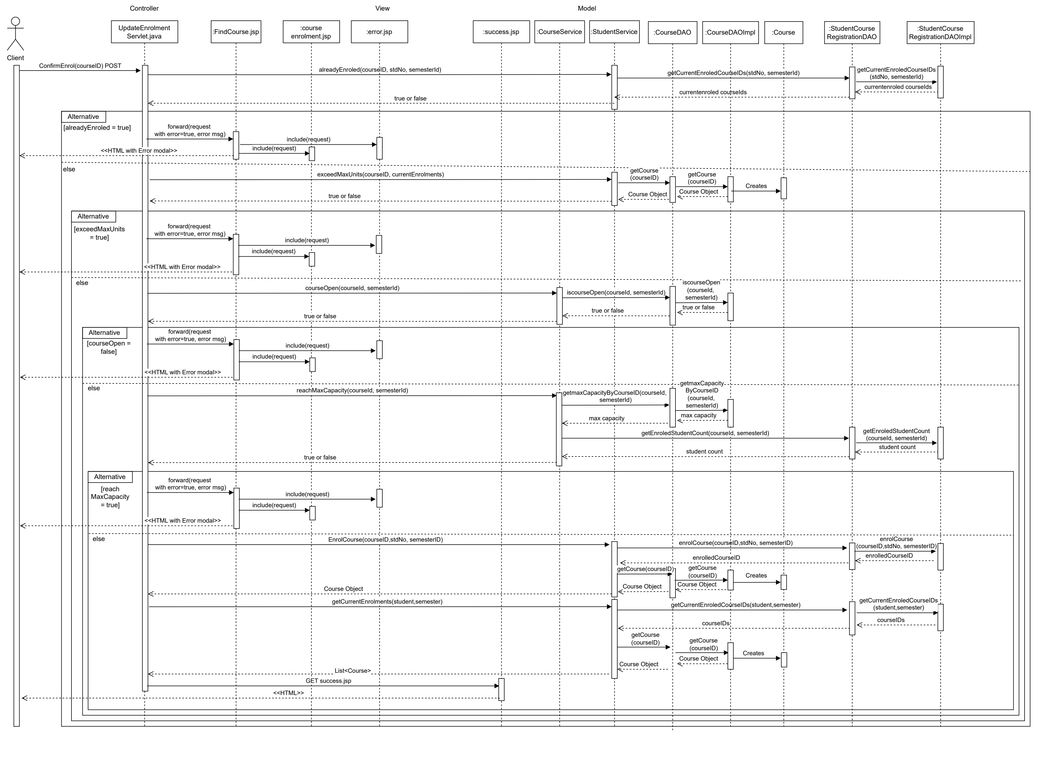

Controller
View
Model
Client
LoginServlet.java
:SemesterSelection Servlet.java
:SemesterSelection.jsp
:index.jsp
:error.jsp
:StudentService
:SemesterService
PasswordSecurity
:StudentDAO
:StudentDAOImpl
:Student
:SemesterDAO
:SemesterDAOImpl
:Semester
Get index.jsp
<<HTML>>
Login(stdNo, password) (POST)
authenticateStudent(stdNo,password)
getStudentByStdNo(stdNo)
getStudentByStdNo (stdNo)
Student Obj or null
Creates
Student Obj or null
verifyPassword(password, student Obj)
true or false
Student Obj (if valid) / null
Alternative
[Object is not null]]
GET SemesterSelection page
getAllSemester
getAllSemester
getAllSemester
List<Semester>
List<Semester>
List<Semester>
Creates
forward(request with semesters = List<Semester>
<<HTML>>
[else]
forward(request with error=true, error msg)
include(request)
modal opens because error=true
<<HTML with Error modal>>


Controller
View
Model
Client
:SemesterSelection Servlet.java
FindCourse Servlet.java
:SemesterSelection.jsp
:FindCourse.jsp
:error.jsp
SemesterService
:StudentService
:SemesterDAO
:SemesterDAOImpl
:Semester
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
:CourseDAO
:CourseDAOImpl
:Course
POST selectSemester(semesterID)
getSemesterBySemesterID(semesterID)
getSemesterBySemesterID(semesterID)
getSemesterBySemesterID (semesterID)
Object
Object
Object
Creates
Alternative
[Semester open for Enrolment]
GET FindCourse page
getCurrentEnrolment(stdNo,semesterID)
getCurrentEnroledCourseIDs(stdNo,semesterID)
getCurrentEnroledCourseIDs (stdNo,semesterID)
courseIDs
courseIDs
getCourseByCourseID(courseID)
getCourseByCourseID (courseID)
Object
Creates
Object
List<Course>
display
<<HTML>>
[else]
forward(request with error=true, error msg)
include(request)
<<HTML with Error modal>>
modal opens because error=true


Controller
View
Model
Client
FindCourse Servlet.java
:FindCourse.jsp
:CourseService
:CourseDAO
:CourseDAOImpl
:Course
searchCourse(coursecode) GET
searchCourse(coursecode)
getAllCourses
getAllCourses
List<Course>
List<Course>
Creates
filtered List<Course>
display search result
<<HTML with course list>>


Controller
View
Model
Client
CourseEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:CourseService
:CourseDAO
:CourseDAOImpl
:Course
GetCourseDetail(courseID) GET
getCourseByCourseID(courseID)
getCourseByCourseID (courseID)
getCourseByCourseID (courseID)
Course Object
Course Object
Creates
Course Object
getAssumedKnowledge(courseID)
getAssumedKnowledge (courseID)
getAssumedKnowledge (courseID)
assumed knowledge courseIDs
assumed knowledge courseIDs
assumed knowledge courseIDs
getPrerequisite(courseID)
getPrerequisite(courseID)
getPrerequisite(courseID)
prerequisite courseIDs
prerequisite courseIDs
prerequisite courseIDs
forward(request with target course, assumed knowledges, prerequisites)
include(request)
<<HTML with course detail sidebar>>
display course detail sidebar including enrol button


Controller
View
Model
Client
CourseEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:error.jsp
:warning.jsp
:confirm.jsp
:StudentService
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
Enrol(courseID) POST
getCompletedCourseIDs(stdNo, semesterId)
getAllEnroledCourseIDs(stdNo)
getAllEnroledCourseIDs(stdNo)
allcourseIDs
allcourseIDs
getCurrentEnroledCourseIDs (stdNo, semesterId)
getCurrentEnroledCourseIDs (stdNo, semesterId)
currentEnroledCourseIds
currentEnroledCourseIds
completedcourseIDs
Alternative
[courseIDs do not include pre-requisite]
forward(request with error=true, error msg)
include(request)
include(request)
<<HTML with Error modal>>
else [courseIDs do not include assumed knowledge course]
forward(request with warning=true, warning msg)
include(request)
include(request)
<<HTML with Warning modal>>
else
forward(request with confirm=true, confirm msg)
include(request)
include(request)
<<HTML with Confirm modal>>


Controller
View
Model
Client
UpdateEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:error.jsp
:success.jsp
:CourseService
:StudentService
:CourseDAO
:CourseDAOImpl
:Course
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
ConfirmEnrol(courseID) POST
alreadyEnroled(courseID, stdNo, semesterId)
getCurrentEnroledCoulseIDs(stdNo, semesterId)
getCurrentEnroledCoulseIDs (stdNo, semesterId)
currentenroled coulseIds
currentenroled coulseIds
true or false
Alternative
[alreadyEnroled = true]
forward(request with error=true, error msg)
include(request)
include(request)
<<HTML with Error modal>>
else
exceedMaxUnits(courseID, currentEnrolments)
getCourse (courseID)
getCourse (courseID)
Course Object
Course Object
Creates
true or false
Alternative
[exceedMaxUnits = true]
forward(request with error=true, error msg)
include(request)
include(request)
<<HTML with Error modal>>
else
courseOpen(courseId, semesterId)
iscourseOpen(courseId, semesterId)
iscourseOpen (courseId, semesterId)
true or false
true or false
true or false
Alternative
[courseOpen = false]
forward(request with error=true, error msg)
include(request)
include(request)
<<HTML with Error modal>>
else
reachMaxCapacity(courseId, semesterId)
getmaxCapacityByCourseID(courseId, semesterId)
getmaxCapacity ByCourseID (courseId, semesterId)
max capacity
max capacity
getEnroledStudentCount(courseId, semesterId)
getEnroledStudentCount (courseId, semesterId)
student count
student count
true or false
Alternative
[reach MaxCapacity = true]
forward(request with error=true, error msg)
include(request)
include(request)
<<HTML with Error modal>>
else
ErrolCourse(courseID,stdNo, semesterID)
errolCourse(courseID,stdNo, semesterID)
errolCourse (courseID,stdNo, semesterID)
errolledCourseID
errolledCourseID
getCourse (courseID)
getCourse (courseID)
Course Object
Course Object
Creates
Course Object
getCurrentEnroledCoulseIDs(student,semester)
getCurrentEnroledCoulseIDs (student,semester)
courseIDs
courseIDs
getCourse (courseID)
getCourse (courseID)
Course Object
Course Object
Creates
List<Course>
GET success.jsp
<<HTML>>